

Huaixiang LI

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EDUCATION

Kunming University of Science and Technology (KUST)

2023 – 2027

Bachelor's Degree in Chemical Engineering and Technology

GPA: 3.63/4.0; Average Score: 87/100

Relevant Courses: Advanced Mathematics, Physical Chemistry, Organic Chemistry, Inorganic and Analytical Chemistry, Principles of Chemical Engineering, Chemical Engineering Thermodynamics, Chemical Process Modelling, Basis of Industrial Catalysis, Instrumental Analysis, Course Design for Chemical Engineering

RESEARCH INTERESTS

Keywords: non-precious metal catalysts, defect engineering, propane dehydrogenation (PDH), CO₂ hydrogenation

- Developing highly efficient non-precious metal catalysts for PDH to address the issues of high cost, carbon deposition, and sintering associated with platinum-based catalysts.
- Investigate catalyst design through defect engineering, alloying, and support modification to tailor active sites and enhance selectivity and stability.
- Extend these strategies to CO₂ valorization, specifically targeting CO₂-assisted propane dehydrogenation (CO₂-ODHP) and CO₂ hydrogenation to methanol, to construct catalysts that efficiently activate inert C-H and CO₂ bonds with long-term durability.

PUBLICATIONS

Tao, Z., **Li, H.**, He, D., Huang, Z., Liang, J., Dong, J., Cao, X., & Luo, Y. (2026). **Ga^{δ+}-H active sites-driven reverse water gas shift reaction with high CO selectivity.** *Chemical Engineering Journal*. [Accepted.]

- First identification of Ga^{δ+}-H species as the key active sites for reverse water-gas shift (RWGS) on Ga/Al₂O₃, enabling rapid CO₂-to-formate transformation.
- Achieves near-100% CO selectivity from 300 °C to 700 °C with exceptional RWGS activity.

Li, H., Zhang, T., Tao, Z., Ling, Y., Zhong, S., Wang, H., Li, Y., He, D., & Luo, Y. (2026). **Stabilizing Ni²⁺ sites via boron-induced structural modulation for propane dehydrogenation.** *Catalysis Science & Technology*. [In peer review.]

- Boron-induced interfacial reconstruction stabilises highly dispersed Ni²⁺ species and suppresses NiO crystallite growth and aggregation.
- Modified electronic structure weakens coke condensation, enhancing reactivity and stability in propane dehydrogenation.

ACADEMIC PROJECTS

High-Efficiency Application and Performance Optimisation of Defect-Controlled Metal Oxide Catalysts in Propane Dehydrogenation

Project Manager | Supervisor: Professor Dedong He

Apr. 2025 – Present

A Funded Project Supported by the Yunnan Provincial University Students' Innovation and Entrepreneurship Fund

- Designed high-performance Ni-based catalysts for PDH through strategic defect engineering and acidity modulation of catalyst supports.
- Led the entire research pipeline, from experimental design and catalyst synthesis to systematic performance evaluation and in-depth data analysis, ensuring the project aligned with its objectives.
- Explored the relationship between the structure and the activity of the catalyst. Mastered and applied advanced characterisation techniques, including XRD, XPS, H₂-TPR, Raman, UV-Vis, EPR, and in-situ DRIFTS, to analyse the catalyst structure, metal valence state, oxygen vacancy defects, and reaction mechanism.
- Achieved exceptional catalytic performance with the developed Ni-based catalyst, maintaining >93% selectivity to propylene and >30% propane conversion at 600°C, surpassing the performance of most Ni-based catalysts reported in the literature.

Silicon-Iron Purification: Waste-to-Green Energy Transformation using Advanced Functional Materials for Water Purification

Core Member | Supervisor: Professor Dedong He

Sept. 2024 – Present

China International University Students' Innovation Competition

- Transformed industrial dangerous waste into high-performance ferrosilicon-based water purification materials, achieving a green circular economy model of treating waste with waste.
- Mastered catalyst material preparation (e.g., ion exchange, impregnation, co-precipitation) and led systematic testing and characterisation to correlate material structure with performance for optimisation.
- Demonstrated strong independent research capabilities in experimental execution and data analysis, building a solid foundation for self-directed experimental and project design.

Research on Environment-Responsive Polysaccharide Nanohydrogels with Hydrophobic Microdomains as a Carrier System

Researcher | Supervisor: Professor Hongying Su

Sept. 2023 – Apr. 2024

A Funded Project Supported by the National Natural Science Foundation of China

- Contributed to a project focusing on developing an intelligent drug delivery system using natural polysaccharides, gaining foundational knowledge in the design of hydrophobic microdomains for insoluble drug loading and stimulus-responsive release mechanisms.
- Established the project's theoretical foundation by conducting a comprehensive literature review on hydrogel design principles and collaborating with the team to optimise the synthesis mechanism.
- Acquired foundational expertise in polysaccharide-based hydrogel systems through laboratory training, which informed collaborative discussions in the catalysis laboratory during the conceptualisation of new experimental proposals.

Preliminary Study on Structural Compatibility between Plant Purple Pigments and Silk/Cotton Fibres

Core Member | Supervisor: Associate Professor Donghui Liu

Apr. 2024 – Apr. 2025

A Funded Project Supported by the Yunnan Provincial University Students' Innovation and Entrepreneurship Fund

- Investigated the colouring mechanism of plant purple pigments on natural fibres by analysing pigment-fibre molecular interactions, aiming to optimise dyeing processes for enhanced colour saturation and fastness.

- Designed and executed a series of dyeing experiments, generating and analysing primary data to identify key factors affecting hue and colour fastness for subsequent mechanistic studies.
- Consolidated the project's theoretical foundation through a literature review on natural pigment dyeing, directly supporting research planning and contributing to a successful patent application.

PROFESSIONAL EXPERIENCE

AI for Science Institute, Beijing

Mar. 2025 – Present

Data Annotator

- Contributed to the training of the Uni-Miner project, which is a large language model. Performed high-quality annotation and proofreading of massive amounts of scientific literature and patent data to build robust training datasets.
- Reviewed the results of the automatic annotations of the AI model, accurately identified and corrected errors, such as entity recognition and relationship extraction, effectively improving the quality and reliability of the dataset.
- Provided critical, structured feedback on recurring patterns of errors. Supported the refinement of the operational protocols of annotation and improved the natural language processing performance of the AI model.

Online Tutoring Initiative

Jan. 2025 – Present

Founder & Tutor

- Leveraged prior tutoring experience to independently launch and manage a private online tutoring service, assuming full responsibility for curriculum development, student acquisition, and session delivery.
- Orchestrated all operational aspects of the initiative, from market outreach and client consultations to scheduling and session delivery, demonstrating strong entrepreneurial initiative.
- Developed a stable client base by developing and implementing long-term, personalised learning schemes for over 15 students, transitioning the service from ad-hoc support to a comprehensive knowledge system building.
- Achieved consistent positive student feedback and sustained business growth, validating the teaching methodology and showcasing effective management and organisational skills.

Liyunlong Education

Jan. 2025 – Feb. 2025

Secondary School Physics & Mathematics Tutor

- Delivered personalised tutoring sessions by developing targeted lesson plans and practice problems, effectively addressing individual learning gaps in physics and mathematics.
- Tracked students' progress through regular assessment. Maintained proactive communication with parents to offer feedback on students' homework and academic performance, facilitating cooperation between the school and the family.

EXTRACURRICULAR ACTIVITIES

Volunteer | United Nations Children's Fund

Feb. 2021 – Present

- Participated in research and advocacy work on the living conditions of children worldwide, gaining a deeper understanding of the core challenges in education and health faced by children across different countries.
- Collaborated within a multicultural team of young volunteers from diverse backgrounds. Gained insights into global social issues and improved cross-cultural communication skills.

Participant | Peking University Guanghua Initiative

Sep. 2022 – Feb. 2023

- Attended a series of structured lectures covering cutting-edge topics and fundamental theories in economics and finance. Applied learnings to explore the crucial role of these two fields in driving social development and resource allocation.
- Reflected on the connection between individual agency and social responsibility; further cultivated abilities to conduct logical analysis and strategic thinking from a macro perspective.

AWARDS AND HONOURS

Competition Awards

- **Third Prize**, the 8th National University Students' Chemical Engineering Experiment Competition (Southwest China Division) **2025**
- **Third Prize**, the Contemporary Undergraduate Mathematical Contest in Modelling (Yunnan Provincial Division) **2024**
- **Third Prize**, the 4th Yunnan Provincial University Students' Energy Conservation & Emission Reduction Competition **2025**
- **Second Prize**, the 18th National University Students' Energy Conservation & Emission Reduction Competition (University Level) **2025**
- **Silver Award**, the China International University Students' Innovation Competition (University Level) **2025**
- **Industry Track Bronze Award**, the China International University Students' Innovation Competition (University Level) **2025**
- **Bronze Award**, the China International University Students' Innovation Competition (University Level) **2024**
- **Third Prize**, the Contemporary Undergraduate Mathematical Contest in Modelling (University Level) **2024**

Scholarships & Honours

- **Provincial "Five Merits" Student** (For Achieving Excellence in Moral, Intellectual, Physical, Aesthetic, and Labour Education) **2026**
- **First-class Scholarship**, KUST **2025**
- **Second-class Scholarship**, KUST **2023 & 2024 & 2026**
- **He Dong Scholarship**, KUST **2024**
- **Innovation and Entrepreneurship Award**, KUST **2025**
- **Merit Student**, KUST **2024**
- **Outstanding Communist Youth League Member**, KUST **2025**

ADDITIONAL INFORMATION

Technical Skills

- **Catalyst Synthesis & Testing:** Ion-exchange, impregnation, co-precipitation; catalytic activity evaluation
- **Characterisation:** H₂-TPR, NH₃-TPD, O₂-TPO, in-situ DRIFTS (CO, reaction, time-resolved), UV-Vis
- **Modelling:** Diamond, C4D

Languages: Chinese (native), English (proficient)